Ricci flow with metric torsion on surfaces of positive Euler characteristic

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Abstract

We study an adapted Ricci flow of connections with metric torsion on surfaces with positive Euler characteristic. We first prove that there do not exist any nontrivial solitons of the flow on the 2-sphere thus confirming a conjecture of Branding–Kröncke [BK17]. We give an explicit family of torsion data for which the corresponding global solutions fail to converge on $\mathbb{S}^2$. Nevertheless, we provide several sufficient conditions for the convergence of the flow to a stationary point. We first prove that the normalized adapted Ricci flow always converges on $\mathbb{RP}^2$, which completely answers a question in [BK17]. Using this, we deduce that the flow converges on $\mathbb{S}^2$ whenever the initial metric and the torsion one-form are antipodally symmetric. We also prove a Łojasiewicz–Simon gradient inequality for the flow and use it to prove convergence to a stationary point provided the solution is close to an arbitrary stationary point.

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1. Introduction

The main purpose of this paper is to study properties of the Ricci flow of metric connections with torsion on two dimensional manifolds with positive Euler characterisitic. The Ricci flow of metric connections with torsion on a Riemann surface was studied by Branding and Kröncke in [BK17] where they studied properties of the flow and used it to prove an analogue of the uniformization theorem for metric connections with torsion when the surface has non-positive Euler characteristic. Let M be a Riemannian surface with a Riemannian metric g. If ∇^{LC} denotes the Levi-Civita connection of g then any affine connection ∇^T on M can be written as

$$\nabla^T_X Y = \nabla^{LC}_X Y + A(X,Y), \qquad X, Y \in \Gamma(TM),$$

where A is a $(2,1)$-tensor field. If in addition, we want the connection ∇^T be a *metric connection*, that is, for all vector fields X, Y, Z we have

$$X(g(Y,Z)) = g(\nabla^T_X Y, Z) + g(Y, \nabla^T_X Z),$$

then A must be skew-adjoint.

Since M is 2-dimensional, there exists a vector field V such that the tensor A takes the form

$$A(X,Y) = g(X,Y)V - g(V,Y)X,$$

and the torsion tensor T of the connection ∇^T is then given by

$$T(X,Y) = A(X,Y) - A(Y,X) = g(V,X)Y - g(V,Y)X.$$

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Throughout the article, we will denote the vectorial torsion with respect to g by

$$V_g = \alpha^{\sharp_g},$$

for some fixed $\alpha \in \Omega^1(M)$. The curvatures of the connection ∇^T have contributions from the tensor T as well. In particular, the scalar curvature R_g^T is

$$R_g^T = R_g + 2\operatorname{div}_g V_g = R_g + 2\operatorname{div}_g \alpha, \tag{1.1}$$

where, by abuse of notation, $\operatorname{div}_g \alpha$ denotes the divergence of the metric-dual vector field $\alpha^{\sharp_g}$. Here R_g is the scalar curvature of the Levi-Civita connection. For a proof of eq. (1.1) and more on geometry of metric connections with vectorial torsion, we refer the reader to [AK16]. We note that if $e^u g$ is a metric conformal to g then the vector field $e^{-u}V$ induces a connetion with the same vectorial torsion.

Motivated by the problem of finding a metric conformal to (M, g) with a given vectorial torsion T or equivalently α such that R_g^T is constant, Branding and Kröncke studied the Ricci flow with metric torsion which we now describe. Let (M^2, g_0) be a closed Riemann surface, $\alpha \in \Omega^1(M)$ be a fixed smooth one-form. The normalized adapted Ricci flow is the following evolution equation for a family of metrics $g(t)$ on M all with fixed vectorial torsion α,

$$\frac{\partial g}{\partial t} = \left(r - R_g^T\right) g, \qquad g(0) = g_0, \tag{1.2}$$

where

$$r = \frac{\int_M R_g^T \, d\mu_g}{\operatorname{Vol}(M, g)} = \frac{4\pi\chi(M)}{\operatorname{Vol}(M, g)}, \tag{1.3}$$

is the average scalar curvature with torsion. The second equality in eq. (1.3) follows from Gauss–Bonnet and the divergence theorem:

$$\int_M R_g^T \, d\mu_g = \int_M R_g \, d\mu_g + 2\int_M \operatorname{div}_g \alpha \, d\mu_g = 4\pi\chi(M).$$

Along the flow eq. (1.2), the volume and hence r are constant. If we write $g = e^u g_0$ for some $u \in C^\infty(M)$ then because the dimension is two, we have

$$R_g = e^{-u}\left(R_{g_0} - \Delta_{g_0} u\right) \qquad \text{and} \qquad \operatorname{div}_g \alpha = e^{-u} \operatorname{div}_{g_0} \alpha, \tag{1.4}$$

where for us $\Delta = \operatorname{div}(\operatorname{grad})$. Consequently,

$$R_g^T = e^{-u}\left(R_{g_0} - \Delta_{g_0} u + 2\operatorname{div}_{g_0} \alpha\right). \tag{1.5}$$

A similar flow in higher dimensions has been studied by Streets [Str08]. The flow is also related to the renormalization group flow in quantum field theory.

Inspired by the proof of the uniformization theorem on surfaces with a Riemannian metric using the Ricci flow in [Ham88], [Cho91], [CLT06], Branding and Kröncke proved the following long-time existence and convergence result for the flow eq. (1.2).

Theorem 1.1 ([BK17], Thm. 1.1)**.** *Let (M, g_0) be a closed Riemannian surface. Then there exists a unique solution $g(t)$ of eq. (1.2) for all $t \in [0, \infty)$. If $\chi(M) \leq 0$ the solution converges to a metric of constant curvature with torsion as $t \to \infty$ and hence in this case, the conformal class of g_0 contains a (up to rescaling) unique metric $\bar{g}$ whose scalar curvature with respect to the torsion α is constant.*

As mentioned in [BK17], the case of the convergence of the flow on surfaces with $\chi(M) > 0$ is a subtle one. In fact, this problem is related to the so-called *prescribed curvature problem* and on $\mathbb{S}^2$ it is known as *Nirenberg's problem*. There have been some outstanding papers on this problem using variational techniques [CY87] as well as parabolic methods [FBR04], [Str05]. One of the important results used in the proof of the uniformization theorem for $\chi(M) > 0$ using the ordinary Ricci flow is that any gradient Ricci soliton on $\mathbb{S}^2$ has constant scalar curvature, a result first proved by Hamilton [Ham88] and also by Chen–Lu–Tian [CLT06], see also [Bre10, Chapter 4]. We first define *self-similar solutions* or *solitons* of the adapted Ricci flow following [BK17].

Definition 1.2. A self-similar solution or a soliton of the normalized adapted Ricci flow eq. (1.2) on a Riemann surface M is a triple (g, α, X), where X is a smooth vector field satisfying

$$\mathscr{L}_X g = (r - R_g^T)g \qquad \text{and} \qquad \mathscr{L}_X \alpha = 0. \tag{1.6}$$

Ths soliton is trivial if

$$R_g^T \equiv r \qquad \text{and} \qquad \mathscr{L}_X g = 0,$$

and it is a gradient soliton if $X = \nabla f$ for some smooth function f.

We note that the first identity in eq. (1.6) implies that X is a conformal vector field on M.

Solitons for eq. (1.2) on $\mathbb{S}^2$ were studied in [BK17] where the authors showed that the round sphere cannot support a nontrivial adapted soliton and moreover, any non-trivial soliton on $\mathbb{S}^2$ can neither be gradient soliton nor can have divergence-free X. The authors conjectured that any soliton on $\mathbb{S}^2$ must be trivial. Our first main result confirms this conjecture.

Theorem 1.3. *Every smooth self-similar solution of the normalized adapted Ricci flow eq. (1.2) on $\mathbb{S}^2$ is trivial.*

As mentioned above, the soliton vector field X is a conformal vector field on $\mathbb{S}^2$. Since all metrics on $\mathbb{S}^2$ determine the standard conformal structure, the flow generated by X is a one-parameter subgroup of $\mathrm{Conf}^+(\mathbb{S}^2) \cong \mathrm{PSL}(2,\mathbb{C})$. Such flows, and equivalently vector fields X, are classified as **elliptic, parabolic** and **hyperbolic.** We show that in the parabolic and the hyperbolic case, the condition $\mathscr{L}_X \alpha = 0$ forces the exact part of α to vanish, and hence $\mathrm{div}_g \alpha = 0$. As a result, the adapted soliton is an ordinary Ricci soliton on $\mathbb{S}^2$ which then is trivial as a result of Kazdan–Warner identity [CK04, Prop. 5.21]. For treating the elliptic case, we look at a Liouville energy eq. (2.1) and prove that it is monotonically decreasing along the flow eq. (1.2). This is proved in Proposition 2.1. We prove Theorem 1.3 in the elliptic case by noticing that the orbits of the flow are periodic and hence the monotonicity formula forces the soliton to be trivial.

It was shown in [BK17, Prop. 4.2] that any staionary point of eq. (1.2) on $\mathbb{S}^2$ except the round metric is linearly unstable. As a result, the authors expected that one cannot expect convergence of the flow on $\mathbb{S}^2$ in general. We confirm this expectation by giving an explicit family of connections with metric torsion such that the global solution to eq. (1.2) exists for all time but fail to converge. More precisely, we prove the following theorem which confirms the expectation in [BK17, Remark 4.3].

Theorem 1.4. *Let $g(t)$ be any solution of eq. (1.2) starting wiht the round metric g_{round} on $\mathbb{S}^2$ with the normalization $r = 2$ and*

$$\alpha = d\left(\frac{1}{2}\log(1 + \varepsilon x_3)\right), \qquad 0 < \varepsilon < 1,$$

where x_3 is the third slot of the Euclidean coordinate function. Then $g(t)$ cannot converge in C^∞ as $t \to \infty$. In fact, it cannot converge in C^2.

The idea to prove this theorem is to prove that if a solution $g(t)$ of the above system converges then it will contradict the Kazdan-Warner identity eq. (2.3) from [KW74], [BE87]. In spite of the above result, we give a sufficient criterion for the convergence on $\mathbb{S}^2$ in Section 3.1.

Proposition 1.5 (Convergence for co-closed torsion)**.** *Let g_0 be any smooth metric on $\mathbb{S}^2$, and let $\alpha \in \Omega^1(\mathbb{S}^2)$ be a smooth one-form satisfying*

$$\mathrm{div}_{g_0} \alpha = 0. \tag{1.7}$$

Let $g(t)$ be the solution to eq. (1.2) with fixed torsion one-form α. Then $g(t)$ exists for all $t \geq 0$ and converges smoothly to a metric g_∞ satisfying $R_{g_\infty}^T \equiv r$.

Even though the previous proposition gives a sufficient condition for convergence of the flow on $\mathbb{S}^2$ and we show that there are infinitely many metric connections with torsion which stisfy the hypothesis of the proposition, the main underlying idea for it to work is that the flow eq. (1.2) does not see the co-closed part of the torsion α. Thus, we seek a better criterion which guarantees convergence. This is the content of Section 3. In Section 3.3 we show that one such condition is antipodal symmetry of the initial metric as well as the torsion one-form α on $\mathbb{S}^2$. Precisely, we prove the following theorem.

Theorem 1.6 (Convergence under antipodal symmetry)**.** *Let g_0 be a smooth metric on $\mathbb{S}^2$ and let $\alpha \in \Omega^1(\mathbb{S}^2)$ be a smooth one-form satisfying*

$$\iota^* g_0 = g_0 \qquad \text{and} \qquad \iota^* \alpha = \alpha, \tag{1.8}$$

where ι is the antipodal map on $\mathbb{S}^2$. Then the normalized adapted Ricci flow with initial metric g_0 and fixed torsion one-form α converges smoothly to a metric g_∞ with constant scalar curvature with torsion

$$R^T_{g_\infty} \equiv r. \tag{1.9}$$

In fact, this result and the condition was obtained when trying to understand the convergence of the flow on $\mathbb{RP}^2$. As explained in [BK17], the situation of convergence of the flow on $\mathbb{RP}^2$ is different than that on $\mathbb{S}^2$ because the former admits metrics which are linearly stable stationary points of the flow unlike, for instance, any metric other than the round metric on $\mathbb{S}^2$. It was mentioned in [BK17] that it is not clear whether one can expect convergence of the flow eq. (1.2) on $\mathbb{RP}^2$ in general. We answer this question completely by proving that the flow converges to a metric with constant scalar curvature with torsion.

Theorem 1.7 (Convergence on $\mathbb{RP}^2$)**.** *Let g_0 be any smooth metric on $\mathbb{RP}^2$ and let $\alpha \in \Omega^1(\mathbb{RP}^2)$ be any smooth one-form. Then the solution $g(t)$ of the normalized adapted Ricci flow eq. (1.2) with initial metric g_0 and fixed torsion one-form α converges smoothly as $t \to \infty$ to a metric g_∞ satisfying*

$$R^T_{g_\infty} \equiv r. \tag{1.10}$$

Before proving this theorem we explain why we get unconditional convergence on $\mathbb{RP}^2$ unlike that on $\mathbb{S}^2$. The difficulty on $\mathbb{S}^2$ is related to the noncompact conformal group of $\mathbb{S}^2$. The idea is that $r_{\mathbb{S}^2}\operatorname{Vol}_{\mathbb{S}^2} = 8\pi$ and $r_{\mathbb{RP}^2}\operatorname{Vol}_{\mathbb{RP}^2} = 4\pi$ which forces an energy-type functional $\mathscr{F}$ in eq. (3.13) from [Cas15] to be coercive on the space of mean-zero functions on $\mathbb{RP}^2$ but not on $\mathbb{S}^2$. As a by-product of our proof of Theorem 1.7, we recover the convergence result of [BK17, Thm. 1.1] for surfaces with $\chi(M) \le 0$.

As mentioned above, one cannot expect convergence of the flow, in general, on $\mathbb{S}^2$. However, we seek to find other conditions which will guarantee convergecne on $\mathbb{S}^2$. We observe in Section 3.2 that the flow eq. (1.2) is the gradient flow of the energy in eq. (2.1) and hence we can hope to apply the general methods of convergence of a flow near its stationary points following Simon [Sim83]. To this end, we prove a Łojasiewicz–Simon gradieent inequality for the flow in Proposition 3.2. Using this, we prove our final conditional convergence for the adapted Ricci flow on $\mathbb{S}^2$. We prove in Theorem 3.16 that if any solution $g(t)$ of the flow is closed in a $W^{2,2}$ neighbourhood of a stationary point $\bar{g}$ (see eq. (3.46) for the precise conditions) then the flow $g(t)$ converges smoothly to a stationary point g_∞ of the flow eq. (1.2).

We note that eq. (1.2), when written in terms of a conformal factor, belongs to the class of mean-field-type flows studied by Castéras. Consequently, the convergence theorem on $\mathbb{RP}^2$ can also be deduced from the subcritical convergence theory in [Cas15], and the antipodally symmetric result on $\mathbb{S}^2$ is related to the equivariant theory in [Cas13]. We nevertheless give a direct proof in the present geometric setting. This proof has several purposes. First, it expresses the compactness mechanism directly in terms of the torsion scalar curvature and the Liouville-type energy of the adapted Ricci flow. Second, it applies without change to nonorientable surfaces, which is important for treating $\mathbb{RP}^2$ intrinsically rather than only through its orientable double cover. Third, it establishes smooth convergence of the evolving metrics by combining coercivity, elliptic and parabolic regularity, and a Łojasiewicz–Simon gradient inequality. Finally, the resulting finite-length and trapping estimates are also used in our conditional convergence theorem near stationary metrics on $\mathbb{S}^2$.

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2. Solitons on $\mathbb{S}^2$ and a nonconvergence result

2.1 Triviality of solitons on $\mathbb{S}^2$

In this section, we prove Theorem 1.3 that any soliton of the normalized adapted Ricci flow on $\mathbb{S}^2$ is trivial, thus confirming the conjecture of [BK17].

We first define the following Liouville type energy for solutions of the normalized adapted Ricci flow on a closed Riemann surface. Let g_0 be metric on a closed Riemann surface M and let $g(t) = e^{u(t)} g_0$ solve the normalized adapted Ricci flow eq. (1.2) with fixed torsion one-form α. Define the Liouville type energy by

$$\mathscr{E}(u) = \int_M \left(\frac{1}{2} |\nabla u|^2_{g_0} + (R_{g_0} + 2 \operatorname{div}_{g_0} \alpha) u - r e^u \right) d\mu_{g_0}. \tag{2.1}$$

We show in Section 3.2 how the flow eq. (1.2) is a gradient flow of the energy. We have the following monotonicity formula.

Proposition 2.1. *Along the adapted Ricci flow $g(t)$, the energy $\mathscr{E}$ satisfies*

$$\frac{d}{dt} \mathscr{E}(u(t)) = - \int_M \left(R^T_{g(t)} - r \right)^2 d\mu_{g(t)}, \tag{2.2}$$

with equality at time t if and only if $R^T \equiv r$ at that time. Thus, the critical points of $\mathscr{E}$ are exactly the metrics with constant curvature with torsion.

Proof. For an arbitrary smooth variation $u + s\eta$, integration by parts gives

$$\begin{aligned} \frac{d}{ds}\Big|_{s=0} \mathscr{E}(u + s\eta) &= \int_M \left(\langle \nabla u, \nabla \eta \rangle_{g_0} + (R_{g_0} + 2 \operatorname{div}_{g_0} \alpha) \eta - r e^u \eta \right) d\mu_{g_0} \\ &= \int_M \left(-\Delta_{g_0} u + R_{g_0} + 2 \operatorname{div}_{g_0} \alpha - r e^u \right) \eta \, d\mu_{g_0}. \end{aligned}$$

By eq. (1.5), we have

$$R_{g_0} + 2 \operatorname{div}_{g_0} \alpha - \Delta_{g_0} u = e^u R^T_g \quad \Longrightarrow \quad -\Delta_{g_0} u + R_{g_0} + 2 \operatorname{div}_{g_0} \alpha - r e^u = e^u (R^T_g - r).$$

Since along the flow we have $\partial_t u = r - R^T_g$, we get

$$\frac{d}{dt} \mathscr{E}(u(t)) = \int_M e^u (R^T_g - r) \partial_t u \, d\mu_{g_0} = - \int_M e^u (R^T_g - r)^2 \, d\mu_{g_0} = - \int_M (R^T_g - r)^2 \, d\mu_g,$$

which proves eq. (2.2). □

We also recall the Kazdan–Warner identity from [KW74], [BE87]. Let g be a smooth Riemannian metric on $\mathbb{S}^2$, and let X be a conformal vector field for g. Then

$$\int_{\mathbb{S}^2} X(R_g) \, d\mu_g = 0. \tag{2.3}$$

Note that If $g = e^u g_{\text{round}}$, where g_{round} is the unit round metric, the conformal vector fields are exactly the real parts of the holomorphic vector fields on the Riemann sphere. In particular, for the gradient conformal fields generated by the coordinate functions x_1, x_2, x_3, identity eq. (2.3) becomes

$$\int_{\mathbb{S}^2} \left\langle \nabla_{g_{\text{round}}} R_g, \nabla_{g_{\text{round}}} x_j \right\rangle_{g_{\text{round}}} e^u \, d\mu_{g_{\text{round}}} = 0. \tag{2.4}$$

If (g, α, X) is a non-trivial soliton on $\mathbb{S}^2$ then from eq. (1.6) we know that X is a conformal vector field. Since every conformal structure on $\mathbb{S}^2$ is the standard one, the flow of X is conjugate to a one-parameter subgroup of the group $\mathrm{PSL}(2, \mathbb{C})$ of Möbius transformations.

We shall use the following standard classification of such vector fields. More about Möbius transformations and properties about elliptic and nonelliptic conformal vector fields can be found in [Bea83, Chapter 4].

Lemma 2.2. *Let $X \not\equiv 0$ be a conformal vector field on $\mathbb{S}^2$. Then exactly one of the following occurs.*

(i) X is of elliptic type. The flow of X, φ_t is periodic which means that there exists $T > 0$ such that $\varphi_T = \mathrm{id}$.

(ii) X is of nonelliptic type. This means that there is a zero p of X such that for every point outside the zero set, either $\varphi_t(q) \to p$ as $t \to +\infty$, or the analogous statement holds after replacing X by $-X$.

Remark 2.3. The nonelliptic vector fields includes the hyperbolic, parabolic, and loxodromic one-parameter Möbius transformations. ▲

We have the following lemma which is essentially [BK17, Lemma 4.4, Prop. 4.6]

Lemma 2.4. *Suppose that X is a conformal vector field on $\mathbb{S}^2$ and $\mathscr{L}_X \alpha = 0$. By the Hodge decomposition*

$$\alpha = d v + \omega, \qquad \operatorname{div}_g \omega = 0. \tag{2.5}$$

Then $X(v) = 0$. In particular, if X is nonelliptic then v is constant and $\operatorname{div}_g \alpha = 0$.

Proof. From $\mathscr{L}_X \alpha = 0$ and eq. (2.5) we obtain

$$d(Xv) + \mathscr{L}_X \omega = 0.$$

The flow of a conformal vector field consists of conformal diffeomorphisms. Since in dimension two, co-closedness of one-forms is conformally invariant hence $\mathscr{L}_X \omega$ is co-closed. It follows that $d(Xv)$ is both exact and co-closed, and therefore

$$\Delta_g (Xv) = 0.$$

Since $\mathbb{S}^2$ is closed we get that the function Xv is constant. Since every conformal vector field on $\mathbb{S}^2$ has a zero we see that this constant is zero. Thus $X(v) = 0$.

Now suppose that X is nonelliptic. By Lemma 2.2, every non-fixed orbit has a limiting fixed point in one time direction. Since v is constant along orbits and is continuous, we get that

$$v(q) = \lim_{t \to +\infty} v(\varphi_t(q)) = v(p).$$

Thus v is constant on $\mathbb{S}^2$. Moreover,

$$\operatorname{div}_g \alpha = \operatorname{div}_g d v = \Delta_g v = 0,$$

which completes the proof of the lemma. □

We now prove the triviality of solitons of the flow eq. (1.2) on $\mathbb{S}^2$.

Proof of Theorem 1.3. If the vector field $X \equiv 0$ then the result holds so lets assume $X \neq 0$. Suppose (g, α, X) is a soliton on $\mathbb{S}^2$. If X is nonelliptic then by Lemma 2.4 $\operatorname{div}_g \alpha = 0$. Hence $R_g^T = R_g$ from eq. (1.1) and the soliton equation becomes

$$(r - R_g) g = \mathscr{L}_X g.$$

Thus g is a two-dimensional Ricci soliton for the ordinary normalized Ricci flow. Thus, using the Kazdan–Warner identity eq. (2.3) we see from [CK04, Prop. 5.21] that the soliton must be trivial.

If X is elliptic then the one-parameter group generated by X is periodic. Thus there exists $T > 0$ such that the flow φ_t of X satisfies

$$\varphi_T = \mathrm{id}_{\mathbb{S}^2}. \tag{2.6}$$

Define $g(t) = \varphi_t^* g$. Since $\mathscr{L}_X \alpha = 0$ we get

$$\varphi_t^* \alpha = \alpha.$$

Since the scalar curvature and the divergence are diffeomorphism invariant hence $R^T_{\varphi_t^* g} = \varphi_t^* R_g^T$. Thus,

$$\frac{\partial g(t)}{\partial t} = \varphi_t^* (\mathscr{L}_X g) = \varphi_t^* \big((r - R_g^T) g \big) = \big(r - R^T_{g(t)} \big) g(t).$$

and as a result $g(t)$ is a solution of the normalized adapted Ricci flow.

Choose a fixed background metric g_0 in the conformal class and write $g(t) = e^{u(t)} g_0$. From Proposition 2.1 we have

$$\frac{d}{dt} \mathscr{E}(u(t)) = - \int_{\mathbb{S}^2} (R^T_{g(t)} - r)^2 \, d\mu_{g(t)}.$$

On the other hand, eq. (2.6) gives $g(T) = \varphi_T^* g = g = g(0)$. Therefore

$$u(T) = u(0) \qquad \text{and} \qquad \mathscr{E}(u(T)) = \mathscr{E}(u(0)).$$

Thus, we have

$$0 = -\int_0^T \int_{\mathbb{S}^2} (R^T_{g(t)} - r)^2 \, d\mu_{g(t)} \, dt, \quad \Longrightarrow \quad R^T_{g(t)} \equiv r \text{ for all } t.$$

In particular,

$$R^T_g \equiv r \qquad \text{and} \qquad \mathscr{L}_X g = 0,$$

which is precisely the triviality of the soliton from Definition 1.2. This completes the proof of the theorem. □

2.2 Non-convergence of the flow on $\mathbb{S}^2$

In this section, we prove Theorem 1.4 by constructing a fixed torsion one-form for which no stationary adapted metric exists on $\mathbb{S}^2$.

Let g_{round} be the unit round metric so that

$$R_{g_{\text{round}}} = 2, \qquad \text{Vol}(\mathbb{S}^2, g_{\text{round}}) = 4\pi, \qquad r = 2.$$

Let

$$x_3 : \mathbb{S}^2 \longrightarrow [-1, 1]$$

be the third slot of the Euclidean coordinate function. We want to use the Kazdan–Warner identity eq. (2.3) and its consequence on the coordinate functions as in eq. (2.4). To this end, choose $0 < \varepsilon < 1$ and define the positive function

$$K = 1 + \varepsilon x_3.$$

Set

$$v = \frac{1}{2} \log K \tag{2.7}$$

and choose the fixed torsion one-form

$$\alpha = dv. \tag{2.8}$$

We have the following observation.

Proposition 2.5. *For the torsion one-form α in eq. (2.8), there is no metric conformal to g_{round} on $\mathbb{S}^2$ having constant scalar curvature with torsion equal to 2.*

Proof. We prove the result by contradiction. Suppose, on the contrary, that $g = e^u g_{\text{round}}$ satisfies

$$R^T_g = 2. \tag{2.9}$$

Using eq. (1.5), $R_{g_{\text{round}}} = 2$, and $\text{div}_{g_{\text{round}}} \alpha = \Delta_{g_{\text{round}}} v$, eq. (2.9) gives

$$2 - \Delta_{g_{\text{round}}}(u - 2v) = 2e^u. \tag{2.10}$$

If we denote by $z = u - 2v$, eq. (2.10) reads as

$$2 - \Delta_{g_{\text{round}}} z = 2e^{z+2v}. \tag{2.11}$$

Consider the conformal metric $\widetilde{g} = e^z g_{\text{round}}$ whose scalar curvature satisfies

$$R_{\widetilde{g}} = e^{-z}\left(2 - \Delta_{g_{\text{round}}} z\right) \overset{(2.11)}{=} R_{\widetilde{g}} = 2e^{2v}.$$

Since scalar curvature is twice the Gaussian curvature in dimension two we have $K_{\tilde{g}} = e^{2v}$ and by the definition of v in eq. (2.7)

$$e^{2v} = K = 1 + \varepsilon x_3. \tag{2.12}$$

Thus the function $1+\varepsilon x_3$ would be the Gaussian curvature of the conformal metric $e^z g_{\text{round}}$ and hence $\nabla_{g_{\text{round}}} K = \varepsilon \nabla_{g_{\text{round}}} x_3$.

Using the consequence eq. (2.4) of the Kazdan–Warner identity for $j = 3$ and for the metric $e^z g_{\text{round}}$ we have

$$\int_{\mathbb{S}^2} \left\langle \nabla_{g_{\text{round}}} K, \nabla_{g_{\text{round}}} x_3 \right\rangle_{g_{\text{round}}} e^z \, d\mu_{g_{\text{round}}} = \varepsilon \int_{\mathbb{S}^2} |\nabla_{g_{\text{round}}} x_3|^2_{g_{\text{round}}} e^z \, d\mu_{g_{\text{round}}} = 0.$$

But $\nabla_{g_{\text{round}}} x_3$ vanishes only at the north and south poles. Hence, as $\varepsilon > 0$, we have

$$\varepsilon \int_{\mathbb{S}^2} |\nabla_{g_{\text{round}}} x_3|^2 e^z \, d\mu_{g_{\text{round}}} > 0,$$

which gives us a contradiction. Therefore no solution of eq. (2.9) exists. □

Proof of Theorem 1.4. Let $g(t)$ be any solution of the flow eq. (1.2) starting with the round metric. Long-time existence of for $g(t)$ follows [BK17, Thm. 1.1]. Let α be the torsion 1-form from eq. (2.8). If we look at the normalized adapted Ricci flow starting from g_{round}, because the flow is conformal, we have $g(t) = e^{u(t)} g_{\text{round}}$ with the conformal factor satisfies

$$\partial_t u = 2 - e^{-u} \left(2 - \Delta_{g_{\text{round}}} u + 2\Delta_{g_{\text{round}}} v\right). \tag{2.13}$$

Suppose, for contradiction, that $u(t) \longrightarrow u_\infty$ in $C^2(\mathbb{S}^2)$. If we define the continuous nonlinear operator

$$\mathcal{N}(u) = 2 - e^{-u} \left(2 - \Delta_{g_{\text{round}}} u + 2\Delta_{g_{\text{round}}} v\right) = \partial_t u.$$

Since $mathcalN : C^2(\mathbb{S}^2) \to C^0(\mathbb{S}^2)$ is continuous, we have

$$\partial_t u(t) = \mathcal{N}(u(t)) \longrightarrow \mathcal{N}(u_\infty) \qquad \text{in} C^0(\mathbb{S}^2).$$

We claim that $\mathcal{N}(u_\infty) = 0$. Suppose not and suppose we have $\mathcal{N}(u_\infty)(p) > 0$ at some $p \in \mathbb{S}^2$. By continuity there exist a neighbourhood $U \ni p$, a constant $c >$ and a time T such that

$$\partial_t u(x,t) \geq c \quad \text{for every } x \in U \text{ and } t \geq T.$$

In particular,

$$u(p,t) - u(p,T) = \int_T^t \partial_t u(p,s) \, ds \geq c(t-T),$$

which contradicts convergence of $u(p,t)$. Thus, $\mathcal{N}(u_\infty) = 0$ which means

$$e^{-u_\infty} \left(2 - \Delta_{g_{\text{round}}} u_\infty + 2\Delta_{g_{\text{round}}} v\right) = 2.$$

But this means that $e^{u_\infty} g_{\text{round}}$ would be a metric of constant scalar curvature with torsion equal to 2. This contradicts Proposition 2.5. Hence the adapted Ricci flow cannot converge in C^2, and therefore cannot converge in C^∞. □

Remark 2.6. Since the Kazdan-Warner identity eq. (2.3) is independent of the initial metric, hence for arbitrary g_0 with $r = \frac{8\pi}{V}$, the same computation proves that the flow fails to converge in C^2 for every initial metric on $\mathbb{S}^2$ by choosing appropriate α depending on the initial metric. ▲

3. Convergence on $\mathbb{S}^2$ and $\mathbb{RP}^2$

3.1 A sufficient condition for convergence on $\mathbb{S}^2$

Although a complete optimal convergence criterion on $\mathbb{S}^2$ is realted to the prescribed Gaussian curvature problem, there is a natural and nontrivial class of torsion one-forms for which convergence follows unconditionally.

Proof of Proposition 1.5. The proof essentially follows from the fact that the divergence is conformally invariant in dimension 2. Since the flow remains in the conformal class of g_0 there is a smooth function $u(t)$ such that $g(t) = e^{u(t)} g_0$. Since the co-differential is $d_g^* \alpha = - *_g d *_g \alpha$ and on one-forms in dimension two, the Hodge star is conformally invariant, whereas the Hodge star on two-forms acquires the factor e^{-u}, we have

$$\delta_{e^u g_0} \alpha = e^{-u} \delta_{g_0} \alpha,$$

and hence

$$\operatorname{div}_{e^u g_0} \alpha = e^{-u} \operatorname{div}_{g_0} \alpha. \tag{3.1}$$

Using eq. (1.7), eq. (3.1) gives $\operatorname{div}_{g(t)} \alpha = 0$ for every $t \geq 0$. Consequently,

$$R^T_{g(t)} = R_{g(t)} + 2 \operatorname{div}_{g(t)} \alpha = R_{g(t)} \implies \frac{\partial g}{\partial t} = (r - R_g) g,$$

which is the ordinary normalized Ricci flow on $\mathbb{S}^2$. The result now follows from classical convergence theorem for the normalized Ricci flow on a closed surface of positive Euler characteristic from [Ham88], [Cho91] and [CLT06] as the limiting metric g_∞ satisfies $R_{g_\infty} \equiv r$, and we have $\operatorname{div}_{g(t)} \alpha = 0$ for all $t \geq 0$ which gives

$$R^T_{g_\infty} = R_{g_\infty} + 2 \operatorname{div}_{g_\infty} \alpha = r.$$

Thus the limiting metric has constant scalar curvature with respect to the prescribed vectorial torsion. □

Remark 3.1. The hypothesis $\operatorname{div}_{g_0} \alpha = 0$ does not require the torsion to vanish. On $\mathbb{S}^2$, since $H^1(\mathbb{S}^2; \mathbb{R}) = 0$, by the Hodge decomposition theorem every co-closed one-form can be written as

$$\alpha = *_{g_0} d v,$$

for some smooth function v. Choosing any nonconstant v gives $\alpha \not\equiv 0$ while still satisfying $\operatorname{div}_{g_0} \alpha = 0$. Thus Proposition 1.5 applies to an infinite-dimensional family of genuinely nonzero torsion tensors. In fact, Proposition 1.5 shows that the normalized adapted Ricci flow with fixed torsion one-form α converges smoothly to a metric g_∞ such that

$$R^T_{g_\infty} \equiv \frac{8\pi}{\operatorname{Vol}(\mathbb{S}^2, g_0)}.$$

▲

3.2 A Łojasiewicz–Simon gradient inequality

We now want to prove other conditional convergence results for solutions on $\mathbb{S}^2$ which remain close to a stationary point of the adapted Ricci flow. We are also interested in analyzing the convergence of the flow to a staionary point on $\mathbb{RP}^2$. Recall from [BK17, Prop. 4.2] that a non-round stationary metric on $\mathbb{S}^2$ is linearly unstable. Consequently, one cannot expect every solution starting close to such a metric to converge to it or any other metric in its diffeomorphism orbit. Nevertheless, the adapted Ricci flow is the gradient flow of the Liouville-type energy introduced in eq. (2.1). Since this energy is real analytic, we prove a version of Łojasiewicz–Simon gradient inequality, following [Sim83], which shows that a solution which remains sufficiently close to a stationary metric cannot oscillate indefinitely. Such a solution has finite length and converges smoothly to a stationary metric.

Let h be a fixed background metric on a closed surface M, let $\alpha \in \Omega^1(M)$ be the fixed torsion one-form and let $g(t) = e^{u(t)} h$. Using eq. (1.5) we denote the operator

$$\mathscr{M}(u) = -\Delta_h u + R^T_h - r e^u = e^u \big(R^T_{e^u h} - r \big), \tag{3.2}$$

and hence

$$\partial_t u = -e^{-u}\mathscr{M}(u). \tag{3.3}$$

Recall also the Liouville-type energy from eq. (2.1)

$$\mathscr{E}(u) = \int_M \left(\frac{1}{2}|\nabla u|_h^2 + R_h^T u - re^u\right) d\mu_h.$$

For every $\eta \in W^{2,2}(M,h)$, its first variation is

$$D\mathscr{E}(u)[\eta] = \int_M \left(\langle \nabla u, \nabla\eta\rangle_h + R_h^T\eta - re^u\eta\right) d\mu_h = \int_M \mathscr{M}(u)\eta\, d\mu_h. \tag{3.4}$$

Thus $\mathscr{M}$ is the $L^2(M,h)$-gradient of $\mathscr{E}$. On the other hand, eq. (3.3) shows that the adapted Ricci flow is the negative gradient flow with respect to the u-dependent inner product

$$\langle \eta, \zeta\rangle_u = \int_M \eta\zeta e^u\, d\mu_h.$$

In particular, as shown in Proposition 2.1

$$\frac{d}{dt}\mathscr{E}(u(t)) = -\int_M (\partial_t u)^2 e^u\, d\mu_h = -\int_M \left(R^T_{g(t)} - r\right)^2 d\mu_{g(t)}. \tag{3.5}$$

We first establish the Łojasiewicz–Simon inequality for any closed Riemann surface which will be used later.

Proposition 3.2 (Łojasiewicz–Simon inequality)**.** *Let (M,h) be a closed Riemannian surface and $\alpha \in \Omega^1(M)$ a fixed one-form. Let $\widetilde{g} = e^{\widetilde{u}}h$ be a smooth stationary metric for the normalized adapted Ricci flow, so that*

$$-\Delta_h\widetilde{u} + R_h^T = re^{\widetilde{u}}. \tag{3.6}$$

Then there exist constants $\sigma > 0$, $C > 0$ and $\theta \in \left(0, \frac{1}{2}\right]$ such that

$$|\mathscr{E}(u) - \mathscr{E}(\widetilde{u})|^{1-\theta} \le C\|\mathscr{M}(u)\|_{L^2(M,h)} \qquad \text{whenever} \qquad \|u - \widetilde{u}\|_{W^{2,2}(M,h)} < \sigma. \tag{3.7}$$

Equivalently, after possibly decreasing σ and enlarging C, we have

$$|\mathscr{E}(u) - \mathscr{E}(\widetilde{u})|^{1-\theta} \le C\left(\int_M \left(R^T_{e^u h} - r\right)^2 d\mu_{e^u h}\right)^{1/2}. \tag{3.8}$$

Proof. We apply the abstract Łojasiewicz–Simon gradient inequality in [Chi03, Thm. 3.10, Cor. 3.11]; see also the original work of Simon [Sim83].

Since M has dimension two, $W^{2,2}(M,h)$ is continuously embedded in $C^0(M)$, and is a Banach algebra, it follows that

$$W^{2,2}(M,h) \longrightarrow W^{2,2}(M,h), \qquad u \longmapsto e^u,$$

is real analytic. Consequently,

$$\mathscr{E} : W^{2,2}(M,h) \longrightarrow \mathbb{R} \qquad \text{and} \qquad \mathscr{M} : W^{2,2}(M,h) \longrightarrow L^2(M,h)$$

are real-analytic maps. Since by eq. (3.6), $\mathscr{M}(\widetilde{u}) = 0$, so $\widetilde{u}$ is a critical point of $\mathscr{E}$. The linearization of $\mathscr{M}$ at $\widetilde{u}$ is

$$D\mathscr{M}(\widetilde{u})[\eta] = -\Delta_h\eta - re^{\widetilde{u}}\eta, \tag{3.9}$$

and the operator

$$D\mathscr{M}(\widetilde{u}) : W^{2,2}(M,h) \longrightarrow L^2(M,h)$$

is a second-order elliptic, self-adjoint Fredholm operator of index zero. Thus all the hypotheses of the abstract Łojasiewicz–Simon inequality from [Chi03, Thm. 3.10] are satisfied, and eq. (3.7) follows.

We now prove the equivalent form eq. (3.8) of the inequality. Again by the Sobolev embedding $W^{2,2}(M,h) \hookrightarrow C^0(M)$, hence after possibly decreasing σ if necessary, there is a constant $A > 0$ such that

$$|u| \le A \qquad \text{whenever} \qquad \|u - \widetilde{u}\|_{W^{2,2}} < \sigma.$$

Using the fact that $\mathscr{M}(u) = e^u\left(R^T_{e^u h} - r\right)$ we obtain

$$\|\mathscr{M}(u)\|^2_{L^2(M,h)} = \int_M e^{2u}\left(R^T_{e^u h} - r\right)^2 d\mu_h = \int_M e^u\left(R^T_{e^u h} - r\right)^2 d\mu_{e^u h}.$$

Since u is uniformly bounded hence e^u is uniformly bounded above and below in the neighbourhood described above, the last expression is uniformly equivalent to $\int_M \left(R^T_{e^u h} - r\right)^2 d\mu_{e^u h}$. This proves eq. (3.8). □

3.3 Convergence on $\mathbb{S}^2$ under antipodal symmetry and on $\mathbb{RP}^2$

Even though Proposition 1.5 gives a sufficient condition for convergence of the flow on $\mathbb{S}^2$, it essentially follows from the simple fact that the flow eq. (1.2) does not see the co-closed part of α through the scalar curvature. We would like to strengthen the criterions for convergence of the flow. The purpose of this subsection is to give another sufficient condition for convergence of the adapted Ricci flow on $\mathbb{S}^2$ which allows the divergence of the torsion one-form to be nonzero.

We first make an obervation that the non-convergent example in Section 2.2 shows that convergence cannot be guaranteed by a universal smallness assumption on α alone. This is because we can simply choose

$$\alpha_\varepsilon = \frac{1}{2} d\log(1+\varepsilon x_3) \longrightarrow 0 \quad \text{in } C^\infty \quad \text{as} \quad \varepsilon \to 0.$$

Indeed, the torsion in that example can be chosen arbitrarily small. The difficulty is instead related to the noncompact conformal group of $\mathbb{S}^2$ and loss of compactness of the solutions. This suggests imposing a symmetry which rules out concentration of energy at a single point.

We prove in this section that the metric and the torsion one-form being antipodal symmetric gives us sufficient condition for convergence on $\mathbb{S}^2$. The reason is that antipodally invariant data on $\mathbb{S}^2$ descend to $\mathbb{RP}^2$. On this quotient the equation is better behaved and for which a convergence theory is available. This is related to the fact that $\mathbb{RP}^2$ admits metrics such that first eigenvalue of the Laplacian satisfy $\lambda_1 > r$ on $\mathbb{RP}^2$, an observation already made in [BK17, Remark 4.3]. Besides giving a nontrivial convergence criterion on $\mathbb{S}^2$, this also answers the question concerning convergence on $\mathbb{RP}^2$ raised in [BK17, Remark 4.3].

We again use a simple variant of the energy in eq. (2.1). Let h be a fixed background metric and let $\alpha \in \Omega^1(M)$ be the fixed torsion one-form. We have

$$R_h^T = R_h + 2\operatorname{div}_h \alpha \qquad \text{and} \qquad \rho = \int_M R_h^T \, d\mu_h = 4\pi\chi(M). \tag{3.10}$$

If $g(t) = e^{u(t)} h$, recall from eq. (1.5) that

$$R^T_{g(t)} = e^{-u(t)}\big(R_h^T - \Delta_h u(t)\big). \tag{3.11}$$

Lemma 3.3. *Let $g(t) = e^{u(t)} h$ be a solution of the normalized adapted Ricci flow with fixed torsion one-form α. Then u satisfies*

$$\frac{\partial}{\partial t} e^u = \Delta_h u - R_h^T + \rho \frac{e^u}{\displaystyle\int_M e^u \, d\mu_h}. \tag{3.12}$$

Moreover, the functional

$$\mathscr{F}(u) = \frac{1}{2}\int_M |\nabla u|_h^2 \, d\mu_h + \int_M R_h^T u \, d\mu_h - \rho \log\left(\int_M e^u \, d\mu_h\right) \tag{3.13}$$

is non-increasing along the flow and satisfies

$$\frac{d}{dt}\mathscr{F}(u(t)) = -\int_M (\partial_t u)^2 e^u \, d\mu_h. \tag{3.14}$$

Proof. Since the normalized adapted Ricci flow preserves the volume, we have

$$\int_M e^{u(t)} \, d\mu_h = \operatorname{Vol}(M, g(t)) = \operatorname{Vol}(M, g(0)) \tag{3.15}$$

for every $t \geq 0$. Multiplying $\partial_t u = r - e^{-u}\big(R_h^T - \Delta_h u\big)$ by e^u gives eq. (3.12). The monotonicity formula in eq. (3.14) follows exactly in the same way as in Proposition 2.1. □

We now explain why the idea of using the functional $\mathscr{F}$ and the Łojasiewicz-Simon inequality can work on $\mathbb{RP}^2$ but not on $\mathbb{S}^2$. Notice that the flow on $\mathbb{S}^2$, for which $\rho = 8\pi$, is critical, whereas on $\mathbb{RP}^2$ one has $\rho = 4\pi < 8\pi$.

We explain heuristically why this distinction is important. Since $\int_M R_h^T \, d\mu_h = \rho$, as explained above the functional $\mathscr{F}$ is invariant under the addition of constants. Indeed, for every $c \in \mathbb{R}$ we have

$$\begin{aligned}\mathscr{F}(u+c) &= \frac{1}{2}\int_M |\nabla u|_h^2 \, d\mu_h + \int_M R_h^T (u+c)\, d\mu_h - \rho \log\left(\int_M e^{u+c}\, d\mu_h\right) \\ &= \frac{1}{2}\int_M |\nabla u|_h^2 \, d\mu_h + \int_M R_h^T u \, d\mu_h + c\rho - \rho c - \rho \log\left(\int_M e^{u}\, d\mu_h\right) \\ &= \mathscr{F}(u).\end{aligned}$$

We may therefore replace u by $u - \overline{u}$, where

$$\overline{u} = \frac{1}{\mathrm{Vol}(M,h)}\int_M u \, d\mu_h,$$

and hence assume that $\overline{u} = 0$. We use the Poincaré inequality

$$\int_M |u - \overline{u}|^2 \, d\mu_h \leq \frac{1}{\lambda_1(h)}\int_M |\nabla u|_h^2 \, d\mu_h, \tag{3.16}$$

where $\lambda_1(h) > 0$ is the first nonzero eigenvalue of $-\Delta_h$. We also use the logarithmic form of the Moser–Trudinger inequality [Mos71], [Cas15, eq. (1.18)]

$$\log\left(\frac{1}{\mathrm{Vol}(M,h)}\int_M e^{u-\overline{u}}\, d\mu_h\right) \leq \frac{1}{16\pi}\int_M |\nabla u|_h^2 \, d\mu_h + C_h, \tag{3.17}$$

where C_h depends only on the background metric h. With the normalization $\overline{u} = 0$, the volume term can be absorbed into the constant, giving

$$\log\left(\int_M e^{u}\, d\mu_h\right) \leq \frac{1}{16\pi}\int_M |\nabla u|_h^2 \, d\mu_h + C_h'. \tag{3.18}$$

Denote by

$$(R_h^T)_0 = R_h^T - \frac{\rho}{\mathrm{Vol}(M,h)} \implies \int_M (R_h^T)_0 \, d\mu_h = 0,$$

which on using $\overline{u} = 0$ gives

$$\int_M R_h^T u \, d\mu_h = \int_M (R_h^T)_0 u \, d\mu_h.$$

We use the Cauchy–Schwarz and Poincaré inequalities to estimate

$$\begin{aligned}\left|\int_M (R_h^T)_0 u \, d\mu_h\right| &\leq \|(R_h^T)_0\|_{L^2(M,h)} \|u\|_{L^2(M,h)} \\ &\leq \lambda_1(h)^{-\frac{1}{2}} \|(R_h^T)_0\|_{L^2(M,h)} \|\nabla u\|_{L^2(M,h)},\end{aligned}$$

and hence by the Young's inequality we get, for every $\varepsilon > 0$,

$$\int_M R_h^T u \, d\mu_h \geq -\varepsilon \int_M |\nabla u|_h^2 \, d\mu_h - C_\varepsilon. \tag{3.19}$$

On the other hand, since $\rho > 0$,eq. (3.18) gives

$$-\rho \log\left(\int_M e^{u}\, d\mu_h\right) \geq -\frac{\rho}{16\pi}\int_M |\nabla u|_h^2 \, d\mu_h - C. \tag{3.20}$$

Thus, combining eq. (3.19) and eq. (3.20), we obtain

$$\mathscr{F}(u) \geq \left(\frac{1}{2} - \frac{\rho}{16\pi} - \varepsilon\right)\int_M |\nabla u|_h^2 \, d\mu_h - C_\varepsilon. \tag{3.21}$$

If $0 < \rho < 8\pi$, then

$$\frac{1}{2} - \frac{\rho}{16\pi} > 0.$$

We may therefore choose $\varepsilon > 0$ sufficiently small so that the coefficient in eq. (3.21) is positive. Consequently, $\mathscr{F}$ is coercive on the space of mean-zero functions. Hence we can expect convergence in the case of $\mathbb{RP}^2$ (and in fact on any closed Riemann surface with $\rho < 8\pi$) but not neccessarily in the case of $\mathbb{S}^2$. Along the flow, $\mathscr{F}$ is non-increasing, and hence $\mathscr{F}(u(t)) \le \mathscr{F}(u(0))$. The coercivity estimate therefore gives a uniform bound for $\|\nabla u(t)\|_{L^2(M,h)}$. The Poincaré inequality then gives a uniform bound for $\|u(t)\|_{L^2(M,h)}$, and consequently

$$\sup_{t\ge 0} \|u(t)\|_{W^{1,2}(M,h)} < \infty.$$

In a nutshell, the decreasing energy controls the $W^{1,2}$-norm of the conformal factor. When $M = \mathbb{S}^2$, the value $\rho = 8\pi$ makes the preceding estimate critical and the coercivity is lost. This is the explanation for concentration along the noncompact conformal group of $\mathbb{S}^2$.

We now turn the coercivity estimate eq. (3.21) into a proof of Theorem 1.7. Rather than appealing to the convergence theory for mean field type flows developed in [Cas15], we give a self-contained argument which uses only eq. (3.21), the maximum principle, interior parabolic estimates, and the Łojasiewicz–Simon inequality of Proposition 3.2. The argument is valid on every closed surface with $\rho < 8\pi$, and in particular recovers, with a different proof, the convergence statement of [BK17, Thm. 1.1] for $\chi(M) \le 0$; see Remark 3.11.

Throughout this discussion (M,h) is an arbitrary closed Riemannian surface, orientable or not, $\alpha \in \Omega^1(M)$ is the fixed torsion one-form, R_h^T and ρ are as in eq. (3.10), and $g(t) = e^{u(t)}h$ is the solution of eq. (1.2) with $g(0) = h$, which exists for all $t \ge 0$ and is smooth on $M \times [0,\infty)$ by [BK17, Thm. 1.1]. By eq. (3.15) the quantities

$$V_0 = \int_M e^{u(t)}\, d\mu_h = \mathrm{Vol}(M, g(t)) \qquad \text{and} \qquad r = \frac{\rho}{V_0} \tag{3.22}$$

are independent of t. We denote the right hand side of eq. (3.14) by

$$\mathscr{D}(t) = \int_M (\partial_t u)^2 e^u\, d\mu_h = \int_M \left(R^T_{g(t)} - r\right)^2 d\mu_{g(t)} = -\frac{d}{dt}\mathscr{F}(u(t)). \tag{3.23}$$

Since $\partial_t e^u = e^u \partial_t u$, eq. (3.12) may be rewritten as the elliptic equation

$$-\Delta_h u = -R_h^T + re^u - e^u \partial_t u, \tag{3.24}$$

whose right-hand side has vanishing h-mean: indeed $\int_M \left(re^u - R_h^T\right) d\mu_h = rV_0 - \rho = 0$ by eq. (3.10) and eq. (3.22), while $\int_M e^u u_t\, d\mu_h = \frac{d}{dt}\mathrm{Vol}(M, g(t)) = 0$. Finally, recall the operator $\mathscr{M}$ from eq. (3.2); the stationary points of eq. (1.2) lying in the conformal class of h and having volume V_0 are exactly the solutions of $\mathscr{M}(u) = 0$ with $\int_M e^u\, d\mu_h = V_0$. We first prove the following *a priori* bound on the evolving conformal factor $u(t)$. Before stating the result and the proof let us make a technical remark. The discussion leading to eq. (3.20) and thereafter uses the fact that $\rho > 0$ whereas all the results are stated (and indeed hold) for $\rho < 8\pi$. In fact, $\rho \le 0$ does not cause any problem and is explained in detail in Remark 3.11.

Lemma 3.4. *Let (M^2, h) be a closed Riemann surface. Suppose $\rho < 8\pi$. Then there exist constants Λ and Λ_p, $1 \le p < \infty$, depending only on h, α and $u(0)$, such that for every $t \ge 0$*

$$\|u(t)\|_{W^{1,2}(M,h)} \le \Lambda \qquad \textit{and} \qquad \left\|e^{u(t)}\right\|_{L^p(M,h)} \le \Lambda_p. \tag{3.25}$$

Moreover $\mathscr{F}$ is bounded from below along the flow, and

$$\int_0^\infty \mathscr{D}(t)\, dt = \mathscr{F}(u(0)) - \mathscr{F}_\infty < \infty, \qquad \mathscr{F}_\infty = \lim_{t\to\infty} \mathscr{F}(u(t)). \tag{3.26}$$

Proof. As observed before eq. (3.16), both $\mathscr{F}$ and $\|\nabla u\|_{L^2(M,h)}$ are invariant under adding constants to u. Hence the estimate eq. (3.21), derived there under the normalization $\overline{u} = 0$, is valid for every $u \in W^{1,2}(M,h)$, that is

$$\mathscr{F}(u) \ge \kappa \int_M |\nabla u|_h^2\, d\mu_h - C_\varepsilon, \qquad \kappa = \frac{1}{2} - \frac{\rho}{16\pi} - \varepsilon > 0,$$

where $\varepsilon > 0$ is fixed once and for all so small that $\kappa > 0$ which is possible precisely because $\rho < 8\pi$. Since $\mathscr{F}(u(t)) \le \mathscr{F}(u(0))$ by eq. (3.14), we obtain

$$\|\nabla u(t)\|^2_{L^2(M,h)} \le \Lambda_1^2 = \kappa^{-1}\left(\mathscr{F}(u(0)) + C_\varepsilon\right). \tag{3.27}$$

We next bound the mean value $\overline{u}(t)$. We have the bbound $\int_M e^{u-\overline{u}}\,d\mu_h \geq \mathrm{Vol}(M,h)$ by Jensen's inequality, while the Moser–Trudinger inequality eq. (3.17) and eq. (3.27) give

$$\int_M e^{u-\overline{u}}\,d\mu_h \leq \mathrm{Vol}(M,h)\exp\left(\frac{\Lambda_1^2}{16\pi}+C_h\right),$$

and hence we have

$$\mathrm{Vol}(M,h) \leq \int_M e^{u-\overline{u}}\,d\mu_h \leq\leq \mathrm{Vol}(M,h)\exp\left(\frac{\Lambda_1^2}{16\pi}+C_h\right).$$

Since $e^{\overline{u}}\int_M e^{u-\overline{u}}\,d\mu_h = V_0$ by eq. (3.22), this yields

$$\frac{V_0}{\mathrm{Vol}(M,h)}\exp\left(-\frac{\Lambda_1^2}{16\pi}-C_h\right) \leq e^{\overline{u}(t)} \leq \frac{V_0}{\mathrm{Vol}(M,h)}, \tag{3.28}$$

and hence $|\overline{u}(t)| \leq C$. Together with eq. (3.27) and the Poincaré inequality eq. (3.16), this proves the first bound in eq. (3.25). Applying eq. (3.17) to pu in place of u gives

$$\log\left(\frac{1}{\mathrm{Vol}(M,h)}\int_M e^{p(u-\overline{u})}\,d\mu_h\right) \leq \frac{p^2}{16\pi}\|\nabla u\|^2_{L^2(M,h)} + C_h \leq \frac{p^2\Lambda_1^2}{16\pi} + C_h,$$

which, combined with eq. (3.28), gives the second bound in eq. (3.25). Finally, coercivity of the functional $\mathscr{F}$ gives $\mathscr{F}(u(t)) \geq -C_\varepsilon$, and $\mathscr{F}(u(t))$ is non-increasing by eq. (3.14); hence it converges to some $\mathscr{F}_\infty \geq -C_\varepsilon$, and integrating eq. (3.14) in time gives eq. (3.26). □

The next lemma is the standard interior regularity theory for eq. (1.2). Note that the only assumption we make is that of a C^0-bound on u.

Lemma 3.5. *Let $A > 0$ and let $g(t) = e^{u(t)}h$ be a solution of eq. (1.2) with $|u| \leq A$ on $M \times [a,b]$, where $b \geq a + \frac{1}{2}$. Then for every integer $k \geq 0$ there is a constant $\Gamma_k = \Gamma_k(A,k,h,\alpha)$, independent of a and b, such that*

$$\sup_{t\in[a+\frac{1}{2},\,b]} \|u(t)\|_{C^k(M)} \leq \Gamma_k. \tag{3.29}$$

Proof. By eq. (3.11) the conformal factor satisfies

$$\partial_t u = e^{-u}\Delta_h u - e^{-u}R_h^T + r. \tag{3.30}$$

Since $e^{-A} \leq e^{-u} \leq e^A$ by $|u| \leq A$, eq. (3.30) is uniformly parabolic with bounded reaction terms on the right-hand side, the ellipticity constants depending only on A. At this stage the coefficient e^{-u} is merely bounded and measurable, so we first apply the interior Hölder estimate of Krylov–Safonov type on the cylinders $M \times [t-\frac{1}{2}, t]$, which gives a uniform $C^{\beta,\frac{\beta}{2}}$-bound for some $\beta \in (0,1)$, see [Lie96, Chapter VII]. The coefficient e^{-u} is then uniformly Hölder continuous in the parabolic metric, so the interior parabolic Schauder estimates apply. Differentiating eq. (3.30) in space and iterating, one obtains eq. (3.29) by the standard bootstrapping argument, see [Lie96, Chapters IV and XII] or [Fri64]. All estimates are interior in time and the cylinders have fixed size, so the constants do not depend on a or b. □

The following elementary comparison estimate controls the flow on the short time intervals on which Lemma 3.5 gives no information. It asserts that the C^0-distance to a stationary metric is controlled, on any fixed time interval, by its value at the initial time of that interval.

Lemma 3.6 (C^0-stability near a stationary metric)**.** *Let $\widetilde{u} \in C^\infty(M)$ satisfy $\mathscr{M}(\widetilde{u}) = 0$, let $g(t) = e^{u(t)}h$ be a solution of eq. (1.2), and set $w = u - \widetilde{u}$. Then*

$$\partial_t w = e^{-u}\Delta_h w + r\big(1 - e^{-w}\big). \tag{3.31}$$

Define

$$\Phi(b,s) = \log\big(1 + (e^b - 1)e^{rs}\big).$$

If $t_0 \geq 0$ and $T > t_0$ are such that $\Phi\big(\sup_M w(t_0), s\big)$ and $\Phi\big(\inf_M w(t_0), s\big)$ are both defined for every $s \in [0, T - t_0]$, that is, both arguments of the logarithm remain positive, then

$$\Phi\big(\inf_M w(t_0), t - t_0\big) \leq w(x,t) \leq \Phi\big(\sup_M w(t_0), t - t_0\big) \qquad \text{for every } (x,t) \in M \times [t_0, T]. \tag{3.32}$$

Consequently, for every $\varepsilon > 0$ and every $\tau > 0$ there is $a = a(\varepsilon, \tau, r) > 0$ such that

$$\|u(t_0) - \widetilde{u}\|_{C^0(M)} \leq a \quad \Longrightarrow \quad \|u(t) - \widetilde{u}\|_{C^0(M)} \leq \varepsilon \quad \text{for every } t \in [t_0, t_0 + \tau]. \tag{3.33}$$

Proof. Since $\mathscr{M}(\widetilde{u}) = 0$ we have $R_h^T - \Delta_h \widetilde{u} = r e^{\widetilde{u}}$, and therefore, by eq. (3.30),

$$\partial_t w = \partial_t u = r - e^{-u}\big(R_h^T - \Delta_h \widetilde{u} - \Delta_h w\big) = r - r e^{\widetilde{u} - u} + e^{-u}\Delta_h w = e^{-u}\Delta_h w + r\big(1 - e^{-w}\big),$$

which is eq. (3.31). We want to apply maximum principle for functions [CK04, Chapter 4]. Let φ solve the ordinary differential equation $\varphi' = r\big(1 - e^{-\varphi}\big)$ with $\varphi(t_0) = \sup_M w(\cdot, t_0) = b_+$. Substituting $y = e^{-\varphi}$ turns this into $y' = r\,y(y-1)$, whose solution with $y(t_0) = e^{-b_+}$ is $y(t_0 + s) = \big(1 + (e^{b_+} - 1)e^{rs}\big)^{-1}$. Hence $\varphi(t_0 + s) = \Phi(b_+, s)$, which by hypothesis is defined on $[0, T - t_0]$. Since φ is constant in space, the function $z = w - \varphi$ satisfies

$$\partial_t z = e^{-u}\Delta_h z + r\big(e^{-\varphi} - e^{-w}\big) = e^{-u}\Delta_h z + b(x,t)\, z,$$

where, by the mean value theorem, $b = r e^{-\xi}$ for some ξ lying between w and φ. The solution u is smooth on $M \times [0, \infty)$, so w is bounded on the compact set $M \times [t_0, T]$, and φ is continuous on $[t_0, T]$, hence b is bounded there. Since $z(\cdot, t_0) \leq 0$ (as $z(\cdot, t_0) = w(\cdot, t_0) - \sup_M w(\cdot, t_0)$), the maximum principle for linear parabolic equations with bounded zeroth-order coefficient gives $z \leq 0$ on $M \times [t_0, T]$, which is the upper bound in eq. (3.32). The lower bound is obtained in the same way, comparing with the solution of the same ordinary differential equation with initial value $\inf_M w(\cdot, t_0)$.

For the last assertion, note first that $\partial_b \Phi(b, s) = e^b e^{rs}\big(1 + (e^b - 1)e^{rs}\big)^{-1} > 0$ wherever Φ is defined, so $b \longmapsto \Phi(b, s)$ is increasing. Notice that $e^{rs} \leq \max\{1, e^{r\tau}\}$ for every $s \in [0, \tau]$, so choose $a > 0$ so small that

$$\big(e^a - 1\big) \cdot \max\{1, e^{r\tau}\} \leq 1 - e^{-\varepsilon}. \tag{3.34}$$

Assume $\|w(t_0)\|_{C^0(M)} \leq a$. For the lower comparison we use $1 - e^{-a} < e^a - 1$, which together with eq. (3.34) gives, for every $s \in [0, \tau]$,

$$1 + \big(e^{-a} - 1\big)e^{rs} = 1 - \big(1 - e^{-a}\big)e^{rs} \geq 1 - \big(e^a - 1\big) \cdot \max\{1, e^{r\tau}\} \geq e^{-\varepsilon} > 0.$$

In particular both comparison functions are defined on $[t_0, t_0 + \tau]$, so eq. (3.32) applies with $T = t_0 + \tau$, and it yields

$$w \geq \Phi(-a, s) \geq \log e^{-\varepsilon} = -\varepsilon.$$

For the upper comparison, eq. (3.34) and $1 - e^{-\varepsilon} \leq e^{\varepsilon} - 1$ give

$$w \leq \Phi(a, s) = \log\big(1 + (e^a - 1)e^{rs}\big) \leq \log\big(1 + (e^a - 1) \cdot \max\{1, e^{r\tau}\}\big) \leq \log\big(2 - e^{-\varepsilon}\big) \leq \varepsilon,$$

the last inequality because $2 \leq e^{\varepsilon} + e^{-\varepsilon}$. This proves eq. (3.33). □

Lemma 3.7 (Subconvergence to a stationary metric)**.** *Let (M, h) be a closed Riemann surface and suppose $\rho < 8\pi$. Then there exist a sequence $t_j \to \infty$ and a function $\widetilde{u} \in C^\infty(M)$ with*

$$\mathscr{M}(\widetilde{u}) = 0 \qquad \text{and} \qquad \int_M e^{\widetilde{u}}\, d\mu_h = V_0, \tag{3.35}$$

such that $u(t_j) \to \widetilde{u}$ in $W^{2,2}(M, h)$ and in $C^0(M)$. Moreover $\mathscr{F}_\infty = \mathscr{F}(\widetilde{u})$, so that

$$H(t) = \mathscr{F}(u(t)) - \mathscr{F}(\widetilde{u}) = \mathscr{E}(u(t)) - \mathscr{E}(\widetilde{u}) \geq 0, \qquad H(t) \longrightarrow 0. \tag{3.36}$$

Proof. By eq. (3.26) we may choose $t_j \in [j, j+1]$ with $\mathscr{D}(t_j) \leq \int_j^{j+1} \mathscr{D}(t)\, dt \to 0$. Hölder's inequality with $\frac{2}{3} = \frac{1}{6} + \frac{1}{2}$, together with eq. (3.25) gives

$$\big\|e^u \partial_t u\big\|_{L^{3/2}(M,h)} \leq \big\|e^{u/2}\big\|_{L^6(M,h)} \big\|e^{u/2}\partial_t u\big\|_{L^2(M,h)} = \big\|e^u\big\|_{L^3(M,h)}^{1/2} \mathscr{D}(t)^{1/2} \leq \Lambda_3^{1/2} \mathscr{D}(t)^{1/2}, \tag{3.37}$$

so the right-hand side of eq. (3.24) is bounded in $L^{3/2}(M,h)$ at $t = t_j$, uniformly in j. As that right-hand side has vanishing mean, elliptic L^p-estimates give $\|u(t_j) - \overline{u}(t_j)\|_{W^{2,3/2}(M,h)} \le C$, while $|\overline{u}(t_j)| \le C$ by eq. (3.28). Since $2 - \frac{2}{3/2} = \frac{2}{3} > 0$, the Sobolev embedding $W^{2,3/2}(M,h) \hookrightarrow C^{0,2/3}(M)$ yields a constant A_1 with $\|u(t_j)\|_{C^0(M)} \le A_1$ for every j.

Using this C^0-bound in place of eq. (3.37) we obtain

$$\left\| e^{u(t_j)} \partial_t u(t_j) \right\|^2_{L^2(M,h)} = \int_M e^{u(t_j)}\, e^{u(t_j)} \partial_t u(t_j)^2 \, d\mu_h \le e^{A_1} \mathscr{D}(t_j) \longrightarrow 0,$$

whence $\|\Delta_h u(t_j)\|_{L^2(M,h)} \le C$ by eq. (3.24), and therefore $\|u(t_j)\|_{W^{2,2}(M,h)} \le C$. Passing to a subsequence we find $\widetilde{u}$ with $u(t_j) \rightharpoonup \widetilde{u}$ weakly in $W^{2,2}(M,h)$ and $u(t_j) \to \widetilde{u}$ in $C^0(M)$, the latter by Rellich-Kondrachov compactness of the embedding $W^{2,2}(M,h) \hookrightarrow C^0(M)$. In particular, $\int_M e^{\widetilde{u}}\, d\mu_h = \lim_j \int_M e^{u(t_j)}\, d\mu_h = V_0$, and by eq. (3.24)

$$\Delta_h u(t_j) = R_h^T - r e^{u(t_j)} + e^{u(t_j)} u_t(t_j) \longrightarrow R_h^T - r e^{\widetilde{u}} \qquad \text{in } L^2(M,h).$$

Since $\Delta_h u(t_j) \rightharpoonup \Delta_h \widetilde{u}$ weakly in $L^2(M,h)$, the limit satisfies $-\Delta_h \widetilde{u} + R_h^T = r e^{\widetilde{u}}$, that is $\mathscr{M}(\widetilde{u}) = 0$, and elliptic regularity together with bootstrapping gives $\widetilde{u} \in C^\infty(M)$. The elliptic estimate

$$\|u(t_j) - \widetilde{u}\|_{W^{2,2}(M,h)} \le C \left(\left\| \Delta_h \big(u(t_j) - \widetilde{u} \big) \right\|_{L^2(M,h)} + \|u(t_j) - \widetilde{u}\|_{L^2(M,h)} \right) \longrightarrow 0$$

upgrades the convergence to strong convergence in $W^{2,2}(M,h)$. Finally $\mathscr{F}$ is continuous on $W^{1,2}(M,h)$ by eq. (3.17), so $\mathscr{F}(u(t_j)) \to \mathscr{F}(\widetilde{u})$, and monotonicity gives $\mathscr{F}_\infty = \mathscr{F}(\widetilde{u})$ and $H \ge 0$. The identity in eq. (3.36) holds because, by Remark 3.14, one has $\mathscr{F} - \mathscr{E} = \rho(1 - \log V_0)$ on the set of all u with $\int_M e^u\, d\mu_h = V_0$, and by eq. (3.35) this set contains $\widetilde{u}$ as well as the whole flow. □

The next two lemmas isolate the two mechanisms which convert the Łojasiewicz–Simon inequality into convergence. Both are used again in Theorem 3.16.

Lemma 3.8 (Łojasiewicz–Simon length estimate)**.** *Let $\widetilde{u} \in C^\infty(M)$ satisfy $\mathscr{M}(\widetilde{u}) = 0$ and let σ, C and θ be the constants of Proposition 3.2. Let $g(t) = e^{u(t)} h$ be a solution of eq. (1.2), let $t_1 < t_2 \le \infty$ and let $A > 0$ be such that, for every $t \in [t_1, t_2)$,*

$$\|u(t) - \widetilde{u}\|_{W^{2,2}(M,h)} \le \sigma, \qquad |u| \le A, \qquad H(t) = \mathscr{E}(u(t)) - \mathscr{E}(\widetilde{u}) \ge 0. \tag{3.38}$$

Then

$$\int_{t_1}^{t_2} \|\partial_t u(t)\|_{L^2(M,h)}\, dt \le \frac{C e^{A/2}}{\theta} H(t_1)^\theta. \tag{3.39}$$

Proof. By eq. (3.5) and eq. (3.23) we have $H'(t) = -\mathscr{D}(t)$, and by eq. (3.8), which is available because $\|u(t) - \widetilde{u}\|_{W^{2,2}(M,h)} \le \sigma$, we have $H(t)^{1-\theta} \le C \mathscr{D}(t)^{1/2}$. Hence, whenever $H(t) > 0$,

$$-\frac{d}{dt} H(t)^\theta = \theta \frac{\mathscr{D}(t)}{H(t)^{1-\theta}} \ge \frac{\theta}{C} \mathscr{D}(t)^{\frac{1}{2}}.$$

If $H(t_0) = 0$ for some $t_0 \in [t_1, t_2)$, then, H being non-negative and non-increasing implies that $H \equiv 0$ and hence $\mathscr{D} \equiv 0$ and $\partial_t u \equiv 0$ on $[t_0, t_2)$; in this case we integrate the above inequality on $[t_1, t_0)$ only. In either case integration gives

$$\int_{t_1}^{t_2} \mathscr{D}(t)^{\frac{1}{2}}\, dt \le \frac{C}{\theta} H(t_1)^\theta.$$

Since $|u| \le A$ implies $\|\partial_t u\|^2_{L^2(M,h)} \le e^A \int_M (\partial_t u)^2 e^u\, d\mu_h = e^A \mathscr{D}$, the estimate eq. (3.39) follows. □

Lemma 3.9 (Smooth convergence from L^2-convergence)**.** *Let $g(t) = e^{u(t)} h$ be a solution of eq. (1.2), let $u_\infty \in C^\infty(M)$ and let $t_1 \ge 0$ be such that*

$$\sup_{t \ge t_1} \|u(t)\|_{C^k(M)} < \infty \quad \text{for every } k \ge 0, \qquad \text{and} \qquad \|u(t) - u_\infty\|_{L^2(M,h)} \longrightarrow 0.$$

Then $u(t) \longrightarrow u_\infty$ in $C^\infty(M)$.

Proof. Fix an integer $k \geq 0$ and choose integers $m > s > k+1$. Interpolation between L^2 and $W^{m,2}$ gives

$$\|u(t) - u_\infty\|_{W^{s,2}(M,h)} \leq C\|u(t) - u_\infty\|_{L^2(M,h)}^{1-\frac{s}{m}} \|u(t) - u_\infty\|_{W^{m,2}(M,h)}^{\frac{s}{m}} \longrightarrow 0,$$

the second factor being bounded by hypothesis. Since $\dim M = 2$ and $s > k+1$, the Sobolev embedding gives $W^{s,2}(M,h) \hookrightarrow C^k(M)$, whence $\|u(t) - u_\infty\|_{C^k(M)} \to 0$. As k was arbitrary, the claim follows. □

We can now prove the convergence theorem. Recall from the Łojasiewicz-Simon gradient inequality Proposition 3.2 that, $\widetilde{u}$ being a smooth stationary point, there are constants $\sigma > 0$, $C > 0$ and $\theta \in \left(0, \frac{1}{2}\right]$ such that, by eq. (3.8) and eq. (3.23),

$$\left|\mathscr{E}(u) - \mathscr{E}(\widetilde{u})\right|^{1-\theta} \leq C\mathscr{D}^{\frac{1}{2}} \qquad \text{whenever} \qquad \|u - \widetilde{u}\|_{W^{2,2}(M,h)} < \sigma. \tag{3.40}$$

Theorem 3.10 (Convergence for $\rho < 8\pi$)**.** *Let (M,h) be a closed Riemannian surface and let $\alpha \in \Omega^1(M)$ be a smooth one-form. If $\rho = 4\pi\chi(M) < 8\pi$, then the solution $g(t)$ of the normalized adapted Ricci flow eq. (1.2) with initial metric h and fixed torsion one-form α converges smoothly as $t \to \infty$ to a metric g_∞ satisfying $R^T_{g_\infty} \equiv r$.*

Proof. Let $\widetilde{u}$, t_j and H be as in Lemma 3.7, and let σ, C and θ be as in eq. (3.40). Put $A = \|\widetilde{u}\|_{C^0(M)} + 1$, let $\Gamma_k = \Gamma_k(A)$ be the constants of Lemma 3.5, let c_S be the norm of the embedding $W^{2,2}(M,h) \hookrightarrow C^0(M)$, let $a > 0$ be the constant provided by eq. (3.33) for $\varepsilon = \frac{1}{2}$ and $\tau = 1$, and set $\sigma' = \min\{\frac{\sigma}{2}, \frac{1}{2c_S}\}$. By the Gagliardo–Nirenberg interpolation inequality on the closed surface M there is a constant C_I with $\|v\|_{W^{2,2}} \leq C_I\|v\|_{L^2}^{1/2}\|v\|_{W^{4,2}}^{1/2}$, and since $\|v\|_{W^{4,2}(M,h)} \leq C\|v\|_{C^4(M)}$ there is $\Theta = \Theta(\Gamma_4, \widetilde{u})$ such that

$$\|u(t) - \widetilde{u}\|_{W^{2,2}(M,h)} \leq \Theta\|u(t) - \widetilde{u}\|_{L^2(M,h)}^{1/2} \qquad \text{whenever } \|u(t)\|_{C^4(M)} \leq \Gamma_4. \tag{3.41}$$

Fix j so large that $\|u(t_j) - \widetilde{u}\|_{C^0(M)} \leq a$, which is possible by Lemma 3.7. Then eq. (3.33) gives $\|u(t) - \widetilde{u}\|_{C^0(M)} \leq \frac{1}{2}$ for every $t \in [t_j, t_j + 1]$, so that $|u| \leq A$ there, and Lemma 3.5 gives $\|u(t)\|_{C^k(M)} \leq \Gamma_k$ for every $t \in [t_j + \frac{1}{2}, t_j + 1]$. Moreover $e^u \geq e^{-A}$ on this cylinder implies $\|u_t\|_{L^2(M,h)} \leq e^{A/2}\mathscr{D}^{1/2}$, so the Cauchy–Schwarz inequality and eq. (3.26) give

$$\left\|u(t_j + \tfrac{1}{2}) - \widetilde{u}\right\|_{L^2(M,h)} \leq \|u(t_j) - \widetilde{u}\|_{L^2(M,h)} + e^{A/2}\left(\int_{t_j}^{t_j+1} \mathscr{D}(t)\,dt\right)^{1/2} \xrightarrow{j\to\infty} 0. \tag{3.42}$$

Since $\|u(t_j + \frac{1}{2})\|_{C^4(M)} \leq \Gamma_4$, eq. (3.41) shows that $\|u(t_j + \frac{1}{2}) - \widetilde{u}\|_{W^{2,2}(M,h)} \to 0$ as $j \to \infty$ and we enlarge j so that in addition we have $\|u(t_j + \frac{1}{2}) - \widetilde{u}\|_{W^{2,2}(M,h)} < \sigma'$. Thus, we moved from t_j to $t_j + \frac{1}{2}$.

Set $s_j = t_j + \frac{1}{2}$ and let

$$T^* = \sup\left\{T \geq s_j \,:\, \|u(t) - \widetilde{u}\|_{W^{2,2}(M,h)} \leq \sigma' \text{ for every } t \in [s_j, T]\right\} \in (s_j, \infty],$$

and assume, for contradiction, that $T^* < \infty$. We will show that T^* is not the supremum in this case. For $t \in [s_j, T^*]$ we have $\|u(t) - \widetilde{u}\|_{C^0(M)} \leq c_S\sigma' \leq 1$, and for $t \in [t_j, s_j]$ we have $\|u(t) - \widetilde{u}\|_{C^0(M)} \leq \frac{1}{2}$ as shown above. Hence $|u| \leq A$ on $M \times [t_j, T^*]$, and Lemma 3.5 gives

$$\|u(t)\|_{C^k(M)} \leq \Gamma_k \qquad \text{for every } t \in [s_j, T^*] \text{ and every } k \geq 0. \tag{3.43}$$

On $[s_j, T^*]$ we have $\|u(t) - \widetilde{u}\|_{W^{2,2}(M,h)} \leq \sigma' \leq \sigma$ and $|u| \leq A$, and $H \geq 0$ by eq. (3.36). Hence Lemma 3.8 applies on $[s_j, T^*]$ and gives

$$\int_{s_j}^{T^*} \|\partial_t u(t)\|_{L^2(M,h)}\,dt \leq \frac{Ce^{A/2}}{\theta} H(s_j)^\theta = L_j, \tag{3.44}$$

and $L_j \to 0$ as $j \to \infty$ by eq. (3.36).

Consequently, for every $t \in [s_j, T^*]$,

$$\|u(t) - \widetilde{u}\|_{L^2(M,h)} \leq \|u(s_j) - \widetilde{u}\|_{L^2(M,h)} + L_j = \delta_j,$$

and $\delta_j \to 0$ by eq. (3.42). Combining this with eq. (3.43) and eq. (3.41) we obtain

$$\|u(t) - \widetilde{u}\|_{W^{2,2}(M,h)} \leq \Theta\delta_j^{1/2} \qquad \text{for every } t \in [s_j, T^*].$$

Enlarging j once more so that $\Theta\delta_j^{1/2} < \frac{\sigma'}{2}$, we conclude that $\|u(T^*) - \widetilde{u}\|_{W^{2,2}(M,h)} < \frac{\sigma'}{2}$. Since $t \longmapsto u(t)$ is continuous with values in $W^{2,2}(M,h)$, the defining inequality of T^* therefore persists on a slightly larger interval, contradicting the maximality of T^*. Hence $T^* = \infty$.

Thus, the estimates eq. (3.43) and eq. (3.44) hold with $T^* = \infty$. In particular $\partial_t u \in L^1\big([s_j,\infty); L^2(M,h)\big)$, so $u(t)$ is Cauchy in $L^2(M,h)$ and converges there to some u_∞. Since $u(t_j) \to \widetilde{u}$ in $L^2(M,h)$, uniqueness of the limit gives $u_\infty = \widetilde{u}$.

By eq. (3.43) and Lemma 3.9, $u(t) \to \widetilde{u}$ in $C^\infty(M)$. Therefore $g(t) = e^{u(t)}h$ converges smoothly to $g_\infty = e^{\widetilde{u}}h$, and $\mathscr{M}(\widetilde{u}) = 0$ means, by eq. (3.2), precisely that $R^T_{g_\infty} \equiv r$. □

We can now easily prove the convergence of the flow to a stationary metric on $\mathbb{RP}^2$.

Proof of Theorem 1.7. Take $h = g_0$ and write $g(t) = e^{u(t)}h$, so that $u(0) = 0$. Since $\chi(\mathbb{RP}^2) = 1$, we have $\rho = 4\pi < 8\pi$, and Theorem 3.10 applies and gives eq. (1.10). □

Remark 3.11. The hypothesis $\rho < 8\pi$ is equivalent to $\chi(M) \le 1$, that is, to M being any closed surface other than $\mathbb{S}^2$. For $\rho \le 0$ the coercivity estimate eq. (3.21) is immediate: with the normalization $\overline{u} = 0$, Jensen's inequality gives $\int_M e^u \, d\mu_h \ge \mathrm{Vol}(M,h)$, whence $-\rho \log\big(\int_M e^u \, d\mu_h\big) \ge -\rho \log \mathrm{Vol}(M,h)$, and the remaining terms are estimated exactly as in eq. (3.19). Consequently Theorem 3.10 contains, in particular, the convergence statement of [BK17, Thm. 1.1] for $\chi(M) \le 0$. The threshold $\rho = 8\pi$ is attained exactly on $\mathbb{S}^2$, where the coercivity is lost, and by Theorem 1.4 the conclusion genuinely fails there. ▲

Remark 3.12. No step of the above argument uses an orientation of M: the Moser–Trudinger inequality eq. (3.17), the Poincaré inequality eq. (3.16), the elliptic and parabolic estimates, and the Łojasiewicz–Simon inequality of Proposition 3.2 are all valid on an arbitrary closed Riemannian surface, with $d\mu_h$ interpreted as the Riemannian density and div_h defined intrinsically. ▲

We can now lift the preceding result to $\mathbb{S}^2$. Let

$$\iota : \mathbb{S}^2 \longrightarrow \mathbb{S}^2, \qquad \iota(x) = -x,$$

be the antipodal map.

Proof of Theorem 1.6. Let $g(t)$ be the solution of the normalized adapted Ricci flow eq. (1.2). We already know from [BK17, Thm. 1.1] that $g(t)$ exists for all $t \ge 0$. By the naturality of scalar curvature and divergence, $\iota^* g(t)$ is also a solution of the same equation with the same fixed torsion one-form α with the initial value $\iota^* g(0) = \iota^* g_0 = g_0$. Uniqueness of the solutions of the adapted flow therefore gives

$$\iota^* g(t) = g(t) \qquad \text{for every } t \ge 0.$$

Thus both $g(t)$ and α descend under the double covering $\pi : \mathbb{S}^2 \longrightarrow \mathbb{RP}^2$.

Denote the descended metric and one-form by $\widehat{g}(t)$ and $\widehat{\alpha}$, respectively. Since the covering map π is a local isometry, $\widehat{g}(t)$ satisfies the normalized adapted Ricci flow on $\mathbb{RP}^2$ with fixed torsion one-form $\widehat{\alpha}$. The normalizing constants agree because both the Euler characteristic and the volume are divided by two under the quotient

$$\frac{4\pi\chi(\mathbb{S}^2)}{\mathrm{Vol}(\mathbb{S}^2, g(t))} = \frac{4\pi\chi(\mathbb{RP}^2)}{\mathrm{Vol}(\mathbb{RP}^2, \widehat{g}(t))}.$$

By Theorem 1.7, $\widehat{g}(t)$ converges smoothly to a metric $\widehat{g}_\infty$ satisfying $R^T_{\widehat{g}_\infty} = r$. Pulling this convergence back by π gives smooth convergence of $g(t)$ to $g_\infty = \pi^* \widehat{g}_\infty$ and proves eq. (1.9). □

Remark 3.13. The condition $\iota^* \alpha = \alpha$ in Theorem 1.6 is a natural geometric condition which ensures that the connection with torsion descends to $\mathbb{RP}^2$. If one is interested only in the evolution of the metric, it is enough to assume

$$\iota^* g_0 = g_0 \qquad \text{and} \qquad \big(\mathrm{div}_{g_0} \alpha\big) \circ \iota = \mathrm{div}_{g_0} \alpha,$$

because as explained before, the metric equation depends on α only through its divergence. The proof by descending the full connection, however, uses the stronger and more geometric hypothesis eq. (1.8). ▲

We finish this section by explaining the relation with the prescribed Gaussian curvature problem and also give explicit examples to which Theorem 1.6 applies but Proposition 1.5 does not. Let g_{round} be the unit round metric, and write the Hodge decomposition of the torsion one-form as

$$\alpha = d v + \omega, \qquad \operatorname{div}_{g_{\text{round}}} \omega = 0. \tag{3.45}$$

If $g = e^u g_{\text{round}}$ and $z = u - 2v$, then

$$R_g^T = e^{-2v} R_{e^z g_{\text{round}}}.$$

Consequently, under the normalization $r = 2$, the stationary equation $R_g^T = 2$ is equivalent to

$$K_{e^z g_{\text{round}}} = e^{2v},$$

K being the Gauss curvature. If $\iota^* \alpha = \alpha$, then v can be chosen antipodally symmetric. Indeed, pullback by ι preserves both the exact and the co-closed summands in eq. (3.45). The uniqueness of the Hodge decomposition therefore gives $d(v \circ \iota - v) = 0$. Applying ι twice shows that the resulting constant is zero. Hence e^{2v} is a positive antipodally symmetric function. We can then apply Moser's theorem [Mos73] or [CY03, Thm. 3] which guarantees that such a function is the Gaussian curvature of an antipodally symmetric conformal metric. In fact, Theorem 1.6 is a dynamical strengthening of this fact: it shows that the adapted Ricci flow converges to such a metric from every antipodally symmetric initial metric.

Finally, for a concrete example, let

$$v = a(3x_3^2 - 1), \qquad \alpha = d v, \qquad a \in \mathbb{R} \setminus \{0\}.$$

Then $v \circ \iota = v$ and $\iota^* \alpha = \alpha$, whereas

$$\operatorname{div}_{g_{\text{round}}} \alpha = \Delta_{g_{\text{round}}} v \not\equiv 0.$$

Thus the adapted flow in this example is not the ordinary Ricci flow, but it nevertheless converges smoothly by Theorem 1.6 but Proposition 1.5 is not applicable in this example. This shows that antipodal symmetry is a genuinely different convergence criterion and applies to examples not covered by Proposition 1.5.

Remark 3.14. The functional $\mathscr{F}$ is used in [Cas15] and is equivalent to the Liouville-type energy $\mathscr{E}$ in eq. (2.1). Indeed, if $V(u) = \int_M e^u \, d\mu_h$ and $\rho = 4\pi\chi(M)$, then the scale-invariant functional is

$$\mathscr{F}(u) = \frac{1}{2} \int_M |\nabla u|^2 \, d\mu_h + \int_M R_h^T u \, d\mu_h - \rho \log V(u).$$

Since the normalized adapted Ricci flow preserves $V(u) = V_0$ and $r = \frac{\rho}{V_0}$, one has

$$\mathscr{F}(u) - \mathscr{E}(u) = \rho(1 - \log V_0),$$

for every u with $\int_M e^u \, d\mu_h = V_0$, in particular along the flow. Hence the two functionals have the same monotonicity formula as shown above and the same critical points on the fixed-volume conformal class. The logarithmic expression is useful because it is invariant under adding constants to u, whereas $\mathscr{E}$ is more directly adapted to the fixed normalization r. ▲

Remark 3.15. Equation (3.12) belongs precisely to the class of flows studied by Castéras in [Cas13; Cas15]. Indeed, upon setting

$$Q = R_h^T, \qquad \rho = \int_M R_h^T \, d\mu_h,$$

equation (3.12) becomes

$$\frac{\partial}{\partial t} e^u = \Delta_h u - Q + \rho \frac{e^u}{\int_M e^u \, d\mu_h},$$

which is exactly the flow considered in [Cas15, eq. (0-4)].

There is also an equivalent formulation which is useful in the presence of symmetries. Let ψ be the unique mean-zero solution of

$$\Delta_h \psi = R_h^T - \frac{\rho}{\operatorname{Vol}(M, h)}$$

and set $f = e^{\psi}$, $u = w + \psi$. Since f is positive and independent of time, equation (3.12) becomes

$$\frac{\partial}{\partial t}(fe^{w}) = \Delta_h w + \rho\left(\frac{fe^{w}}{\int_M fe^{w}\,d\mu_h} - \frac{1}{\mathrm{Vol}(M,h)}\right).$$

This is the equivariant flow considered in [Cas13]. Consequently, the subcritical convergence result [Cas15, Theorem 0.2(i)] applies when $\rho < 8\pi$, while the equivariant convergence theorem [Cas13, Theorem 1.2] applies when the data are invariant under an isometry group G satisfying

$$|G \cdot x| > \frac{\rho}{8\pi} \qquad \text{for every } x \in M.$$

Thus Theorem 1.7 can alternatively be deduced from the subcritical convergence theorem [Cas15, Theorem 0.2(i)], while Theorem 1.6 follows from the equivariant convergence theorem [Cas13, Theorem 1.2]. We retain a direct proof for three reasons: it identifies explicitly the geometric coercivity mechanism for the adapted Ricci flow, treats nonorientable surfaces intrinsically, and provides the smooth compactness and finite-length estimates used later in the local convergence analysis. Accordingly, the novelty of Theorems 1.6 and 1.7 lies in their formulation and geometric interpretation for metric connections with vectorial torsion, rather than in a new general subcritical mean-field convergence theorem ▲

3.4 Another conditional convergence on $\mathbb{S}^2$

We finally prove our last conditional convergence theorem.

Theorem 3.16 (Conditional convergence near a stationary metric)**.** *Let $\widetilde{g} = e^{\widetilde{u}}h$ be a smooth stationary metric for the normalized adapted Ricci flow eq. (1.2) on $\mathbb{S}^2$. There exists $\delta > 0$ with the following property. Let $g(t) = e^{u(t)}h$ be a solution of the normalized adapted Ricci flow. Suppose that there exists $T \geq 0$ such that*

$$\|u(t) - \widetilde{u}\|_{W^{2,2}(\mathbb{S}^2,h)} < \delta \qquad \text{for every } t \geq T. \tag{3.46}$$

Then there exists a smooth function u_∞ such that

$$u(t) \longrightarrow u_\infty \qquad \text{in} \quad C^\infty(\mathbb{S}^2) \quad \text{as } t \to \infty.$$

The limiting metric $g_\infty = e^{u_\infty}h$ is stationary, meaning, $R^T_{g_\infty} \equiv r$. Moreover,

$$\mathscr{E}(u_\infty) = \mathscr{E}(\widetilde{u}). \tag{3.47}$$

If $\widetilde{u}$ is isolated among the solutions of eq. (3.6) in a sufficiently small $W^{2,2}$-neighbourhood, then $u_\infty = \widetilde{u}$. The same conclusion holds if there exists a sequence $t_j \to \infty$ such that $u(t_j) \longrightarrow \widetilde{u}$ in $W^{2,2}(\mathbb{S}^2,h)$.

Proof. Let σ, C and θ be the constants given by Proposition 3.2, let c_S be the norm of the embedding $W^{2,2}(\mathbb{S}^2,h) \hookrightarrow C^0(\mathbb{S}^2)$, and set

$$\delta = \sigma, \qquad A = \|\widetilde{u}\|_{C^0(\mathbb{S}^2)} + c_S\delta,$$

with the possibility of shrinking δ in the isolated case. By eq. (3.46) we have $|u| \leq A$ on $\mathbb{S}^2 \times [T,\infty)$. Applying Lemma 3.5 on $\mathbb{S}^2 \times [T,b]$ for every $b \geq T + \frac{1}{2}$ we obtain constants $\Gamma_k = \Gamma_k(A)$, independent of b, with

$$\|u(t)\|_{C^k(\mathbb{S}^2)} \leq \Gamma_k \qquad \text{for every } t \geq T + \tfrac{1}{2} \text{ and every } k \geq 0. \tag{3.48}$$

In particular $\{u(t) : t \geq T + \frac{1}{2}\}$ is precompact in $W^{2,2}(\mathbb{S}^2,h)$.

We now prove subconvergence to a stationary metric. By eq. (3.46) the quantities $\|u(t)\|_{C^0(\mathbb{S}^2)}$ and $\|\nabla u(t)\|_{L^2(\mathbb{S}^2,h)}$ are bounded uniformly in $t \geq T$, so $\mathscr{E}(u(t))$ is bounded; being non-increasing by Proposition 2.1, it converges to some $\mathscr{E}_\infty \in \mathbb{R}$, and integrating eq. (3.5) gives

$$\int_T^\infty \mathscr{D}(t)\,dt = \mathscr{E}(u(T)) - \mathscr{E}_\infty < \infty.$$

Hence there is a sequence $t_j \to \infty$ with $\mathscr{D}(t_j) \to 0$. Since $|u| \leq A$,

$$\|\mathscr{M}(u(t_j))\|^2_{L^2(\mathbb{S}^2,h)} = \int_{\mathbb{S}^2} e^{u(t_j)}\left(R^T_{g(t_j)} - r\right)^2 d\mu_{g(t_j)} \leq e^A\mathscr{D}(t_j) \longrightarrow 0,$$

where we used eq. (3.2) and eq. (3.23). By the precompactness established above we may pass to a subsequence and find $u_\infty \in W^{2,2}(\mathbb{S}^2, h)$ with

$$u(t_j) \longrightarrow u_\infty \qquad \text{in } W^{2,2}(\mathbb{S}^2, h). \tag{3.49}$$

As $\mathscr{M} : W^{2,2}(\mathbb{S}^2, h) \to L^2(\mathbb{S}^2, h)$ is continuous, $\mathscr{M}(u_\infty) = 0$, that is $-\Delta_h u_\infty + R_h^T = r e^{u_\infty}$ weakly, hence, elliptic regularity and bootstrapping give $u_\infty \in C^\infty(\mathbb{S}^2)$ and $R^T_{e^{u_\infty} h} \equiv r$.

Since $\delta \le \sigma$, the inequality eq. (3.8) holds along the trajectory for every $t \ge T$; evaluating it at $t = t_j$ and using eq. (3.23) gives

$$\left|\mathscr{E}(u(t_j)) - \mathscr{E}(\widetilde{u})\right|^{1-\theta} \le C \mathscr{D}(t_j)^{\frac{1}{2}} \longrightarrow 0,$$

whence $\mathscr{E}_\infty = \mathscr{E}(\widetilde{u})$. As $\mathscr{E}(u(t))$ is non-increasing with limit $\mathscr{E}(\widetilde{u})$, the function $H(t) = \mathscr{E}(u(t)) - \mathscr{E}(\widetilde{u})$ satisfies $H \ge 0$ and $H \to 0$ on $[T, \infty)$. Letting $j \to \infty$ in eq. (3.49) and using the continuity of $\mathscr{E}$ we obtain eq. (3.47).

The hypotheses eq. (3.38) of Lemma 3.8 now hold on $[T, \infty)$, so

$$\int_T^\infty \|\partial_t u(t)\|_{L^2(\mathbb{S}^2, h)}\, dt \le \frac{C e^{A/2}}{\theta} H(T)^\theta < \infty.$$

Therefore $u(t)$ is Cauchy in $L^2(\mathbb{S}^2, h)$ and converges there; by eq. (3.49) the limit is u_∞. Combining eq. (3.48) with Lemma 3.9 gives $u(t) \to u_\infty$ in $C^\infty(\mathbb{S}^2)$, and $g_\infty = e^{u_\infty} h$ is stationary as shown above.

If $\widetilde{u}$ is isolated among the stationary points in the chosen neighbourhood, then choose an isolation radius $\varepsilon_{\text{iso}} > 0$, and choose

$$\delta < \min\{\sigma, \varepsilon_{\text{iso}}\}.$$

Since $u(t)$ remains within δ of $\widetilde{u}$ and $u(t) \to u_\infty$ in $W^{2,2}$, one has

$$\|u_\infty - \widetilde{u}\|_{W^{2,2}} \le \delta < \varepsilon_{\text{iso}}.$$

As u_∞ is stationary, isolation then gives $u_\infty = \widetilde{u}$. Likewise, if $u(t_j) \to \widetilde{u}$ along some sequence, then the uniqueness of the L^2-limit implies directly that $u_\infty = \widetilde{u}$. □

Remark 3.17. As is evident from the proof, Theorem 3.16 s true for any closed Riemann surface M, but the most interestng case is that of $M = \mathbb{S}^2$ as for other cases we have an unconditional convergence theorem. ▲

Corollary 3.18 (The nondegenerate case)**.** *Let $\widetilde{g} = e^{\widetilde{u}} h$ be a stationary metric and suppose that*

$$r \notin \operatorname{Spec}(-\Delta_{\widetilde{g}}). \tag{3.50}$$

Then $\widetilde{u}$ is an isolated stationary point. Moreover, the exponent in Proposition 3.2 may be chosen to be $\theta = \frac{1}{2}$. Consequently, any solution satisfying the assumption eq. (3.46) converges exponentially in C^∞ to $\widetilde{u}$.

Proof. Multiplying eq. (3.9) by $e^{-\widetilde{u}}$ gives

$$e^{-\widetilde{u}} D\mathscr{M}(\widetilde{u})[\eta] = -\Delta_{\widetilde{g}} \eta - r\eta.$$

Thus

$$\ker D\mathscr{M}(\widetilde{u}) = \left\{\eta \in W^{2,2}(\mathbb{S}^2) \mid -\Delta_{\widetilde{g}} \eta = r\eta\right\}.$$

As a result, assumption eq. (3.50) implies that $D\mathscr{M}(\widetilde{u})$ is invertible. The Banach space implicit function theorem shows that $\widetilde{u}$ is an isolated stationary point. For a nondegenerate critical point, the Łojasiewicz–Simon exponent can be chosen to be $\theta = \frac{1}{2}$.

Using the same notation as in the proof of Theorem 3.16, we have $H(t)^{1/2} \le C\mathscr{D}(t)^{1/2}$ and $H'(t) = -\mathscr{D}(t) \le -C^{-2} H(t)$, so that $H(t) \le C e^{-ct}$. Lemma 3.8 applied on $[t, \infty)$ then gives $\|u(t) - \widetilde{u}\|_{L^2(\mathbb{S}^2, h)} \le C' e^{-\frac{ct}{2}}$, and eq. (3.48) together with the interpolation in the proof of Lemma 3.9 upgrades this to exponential convergence in every C^k-norm. We mention that this result excludes the round metric on $\mathbb{S}^2$ as the condition in eq. (3.50) fails. □

Remark 3.19. The assumption eq. (3.46) in Theorem 3.16 cannot in general be replaced by the assumption that $u(0)$ is merely close to $\widetilde{u}$. In fact, as shown in [BK17, Prop. 4.2] the linearization of eq. (1.2) at a stationary metric is

$$\eta_t = \Delta_{\widetilde{g}}\eta + r\eta.$$

Thus an eigenfunction satisfying $-\Delta_{\widetilde{g}}\eta = \lambda\eta$ has linear growth rate $r - \lambda$ and hence on $\mathbb{S}^2$, Hersch's inequality [Her70] gives

$$\lambda_1(\widetilde{g}) \leq \frac{8\pi}{\mathrm{Vol}(\mathbb{S}^2, \widetilde{g})} = r,$$

with equality only for a round metric. Consequently, every non-round stationary metric has an unstable direction. What Theorem 3.16 asserts is that trajectories of flows which do not escape along such an unstable direction nevertheless converge to a stationary metric. ▲

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